Description:

Suppose that a player walks into a casino with $5 in their pocket and plays 5 hands of a game, say Blackjack, where the player pays $1 to play a hand and their probability of winning the $2 reward in a hand is 0.30. In this scenario, our random variable of interest is the amount of money left in the player's pocket after they have played all five hands. Let this random variable be denoted by *X*. Investigate *X* by:

1. Simulate 1000 random values for X.
2. What is the theoretical probability distribution of X? What is E(X)? What is VAR(X)?
3. Compare the simulated frequency distribution of X with its theoretical distribution by using the chi-square goodness-of-fit-test.